THE INFINITESIMAL LIFTING CRITERION FOR SMOOTH MORPHISMS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4586-4596 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 4598-4693, the complete author proof (94 body lines).

Editorial note. Navigation stubs: 18 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Formally smooth morphisms of algebraic spaces, etale charts, and the cohomological obstruction to smoothness; the implication is retained.

Source reference. Original lemma coherent-lemma-quasi-coherent-affine-cohomology-zero: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4693–4693. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma definition-formally-smooth: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4491–4496. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma descent-proposition-same-cohomology-quasi-coherent: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4693–4693. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-action-sheaf: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3827–3838. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-composition-formally-smooth-etale-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2486–2495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-etale-on-top: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2567–2576. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-finite-presentation-differentials: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1265–1270. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-helper-formally-smooth: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4554–4565. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-representable-property-formally-property: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2529–2537. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-sheaf: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3791–3805. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-lemma-smooth-formally-smooth: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4693–4693. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma remark-special-case: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3848–3864. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma sites-cohomology-definition-torsor: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4693–4693. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma sites-cohomology-lemma-torsors-h1: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4693–4693. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-finite-presentation-local: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4693–4693. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-smooth-local: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4693–4693. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-properties-lemma-properties-quasi-coherent: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4693–4693. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-properties-section-properties-modules: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4693–4693. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1 (Infinitesimal lifting criterion)**.** *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The following are equivalent:*

(1) *The morphism f is smooth.*
(2) *The morphism f is locally of finite presentation, and formally smooth.*

Proof. Assume $f : X \to S$ is locally of finite presentation and formally smooth. Consider a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{\psi} & V \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

where U and V are schemes and the vertical arrows are étale and surjective. By Lemma lemma-helper-formally-smooth we see $\psi : U \to V$ is formally smooth. By Morphisms of Spaces, Lemma spaces-morphisms-lemma-finite-presentation-local the morphism ψ is locally of finite presentation. Hence by the case of schemes the morphism ψ is smooth, see More on Morphisms, Lemma more-morphisms-lemma-smooth-formally-smooth. Hence f is smooth, see Morphisms of Spaces, Lemma spaces-morphisms-lemma-smooth-local.

Conversely, assume that $f : X \to Y$ is smooth. Consider a solid commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{a} & T \\ {\scriptstyle f}\downarrow & \nwarrow & \downarrow{\scriptstyle i} \\ Y & \xleftarrow{} & T' \end{array}$$

as in Definition definition-formally-smooth. We will show the dotted arrow exists thereby proving that f is formally smooth. Let $\mathcal{F}$ be the sheaf of sets on $(T')_{spaces,étale}$ of Lemma lemma-sheaf as in the special case discussed in Remark remark-special-case. Let

$$\mathcal{H} = \mathcal{H}om_{\mathcal{O}_T}(a^*\Omega_{X/Y}, \mathcal{C}_{T/T'})$$

be the sheaf of $\mathcal{O}_T$-modules on $T_{spaces,étale}$ with action $\mathcal{H} \times \mathcal{F} \to \mathcal{F}$ as in Lemma lemma-action-sheaf. The action $\mathcal{H} \times \mathcal{F} \to \mathcal{F}$ turns $\mathcal{F}$ into a pseudo $\mathcal{H}$-torsor, see Cohomology on Sites, Definition sites-cohomology-definition-torsor. Our goal is to show that $\mathcal{F}$ is a trivial $\mathcal{H}$-torsor. There are two steps: (I) To show that $\mathcal{F}$ is a torsor we have to show that $\mathcal{F}$ has étale locally a section. (II) To show that $\mathcal{F}$ is the trivial torsor it suffices to show that $H^1(T_{étale}, \mathcal{H}) = 0$, see Cohomology on Sites, Lemma sites-cohomology-lemma-torsors-h1.

First we prove (I). To see this choose a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{\psi} & V \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

where U and V are schemes and the vertical arrows are étale and surjective. As f is assumed smooth we see that ψ is smooth and hence formally smooth by Lemma lemma-representable-property-formally-property. By the same lemma the morphism $V \to Y$ is formally étale. Thus by Lemma lemma-composition-formally-smooth-etale-unramified the composition $U \to Y$ is formally smooth. Then (I) follows from Lemma lemma-etale-on-top part (4).

Finally we prove (II). By Lemma lemma-finite-presentation-differentials we see that $\Omega_{X/S}$ is of finite presentation. Hence $a^*\Omega_{X/S}$ is of finite presentation (see Properties of Spaces, Section spaces-properties-section-properties-modules). Hence the sheaf $\mathcal{H} = \mathcal{H}om_{\mathcal{O}_T}(a^*\Omega_{X/Y}, \mathcal{C}_{T/T'})$ is quasi-coherent by Properties of Spaces, Lemma spaces-properties-lemma-properties-quasi-coherent. Thus by Descent, Proposition descent-proposition-same-cohomology-quasi-coherent and Cohomology of Schemes, Lemma coherent-lemma-quasi-coherent-affine-cohomology-zero we have

$$H^1(T_{spaces,étale}, \mathcal{H}) = H^1(T_{étale}, \mathcal{H}) = H^1(T, \mathcal{H}) = 0$$

as desired. □